

Exercise: Text Analyzer

Grading Rubric: Text Analyzer

Functionality

count_statistics (3 points)

- **1 point:** Function correctly counts the number of words
- **1 point:** Function correctly counts the number of sentences (using ., !, ?)
- **1 point:** Function correctly counts characters excluding spaces

word_frequency (5 points)

- **1 point:** Function correctly counts word frequencies
- **1 point:** Results are sorted by frequency in descending order
- **1 point:** Implementation correctly handles the `ignore_stopwords` parameter
- **1 point:** Words are compared case-insensitively (e.g., “The” and “the” count as same word)
- **1 point:** Punctuation is properly removed before counting

evaluate_readability (4 points)

- **1 point:** Function correctly evaluates average word length
- **1 point:** Function correctly evaluates average sentence length
- **1 point:** Score calculation is reasonable and consistent
- **1 point:** Feedback is appropriate and mentions relevant text characteristics

Code Quality (5 points)

- **1 point:** Code is well-organized and follows Python style conventions
- **1 point:** Variable names are clear and descriptive
- **1 point:** Docstrings are complete and accurately describe functions
- **1 point:** Code includes comments where appropriate to explain complex logic
- **1 point:** Code handles edge cases properly (e.g., empty text, text without punctuation)

Testing (5 points)

- **5 points:** All unit tests pass

Total: 22 points

Bonus Points (up to 3 extra points)

- **2 points:** Add a function `extract_key_sentences(text, n=3)` for extractive summarization based on word frequency
- **1 point:** Implement a function `detect_language(text)` that guesses the language based on stopwords frequency